

**University of Essex Online – MSc Cybersecurity**

# **Cloud Infrastructure and Framework Report**

**Group C – Group Project**

**Module: Cloud Operations & Management**

**Unit 6**

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# 1. Overview

Azure Resource Manager (ARM) is Microsoft Azure's deployment and management service. ARM provides a management layer for the consistent creation, update and deletion of cloud resources. It supports Infrastructure as Code (IaC) using JavaScript Object Notation (JSON)-based ARM templates. Bicep is Microsoft's declarative, domain-specific language that provides concise syntax, type safety and code reusability, and it is converted into ARM templates during deployment (Microsoft, 2026a; Microsoft, 2026b).

ARM enables consistent, repeatable infrastructure deployment across development, testing and production environments. Its dependency management reduces deployment-order errors when dependencies are correctly defined in the infrastructure code. Azure provides four management scopes: management groups, subscriptions, resource groups and resources, enabling governance and access controls to be applied hierarchically (Microsoft, 2026a).

Bicep complements ARM through modular infrastructure design, enabling reusable components and dependencies within infrastructure code (Microsoft, 2026c). What-If previews changes before deployment, reducing unintended changes and supporting change control (Microsoft, 2026d).

# 2. Service Model Analysis

ARM and Bicep are not cloud service models; rather, they provide technologies for automating the management of Azure resources used within the IaaS, PaaS and SaaS models.

Under Infrastructure as a Service (IaaS), ARM can provision resources such as virtual machines, virtual networks and load balancers. This provides greater infrastructure control, but customers remain responsible for the operating system and applications, unlike the more highly managed PaaS and SaaS models (Microsoft, 2026e).

Under the Platform as a Service (PaaS) model, ARM can provision and configure managed services including Azure App Service, Azure SQL Database, Azure Functions and Azure Container Apps. ARM automates application-platform deployment and configuration while Azure

manages much of the underlying infrastructure. This reduces infrastructure administration (Microsoft, 2026e).

Under Software as a Service (SaaS), ARM does not provide the application's business functionality; it primarily supports Azure resource deployment and management. Responsibility also shifts across these models: Microsoft assumes more responsibility in SaaS than PaaS, while customers retain greater control and responsibility under IaaS (Microsoft, 2026e).

Overall, ARM and Bicep enable governed, repeatable Azure deployments across IaaS, PaaS and supporting SaaS resources.

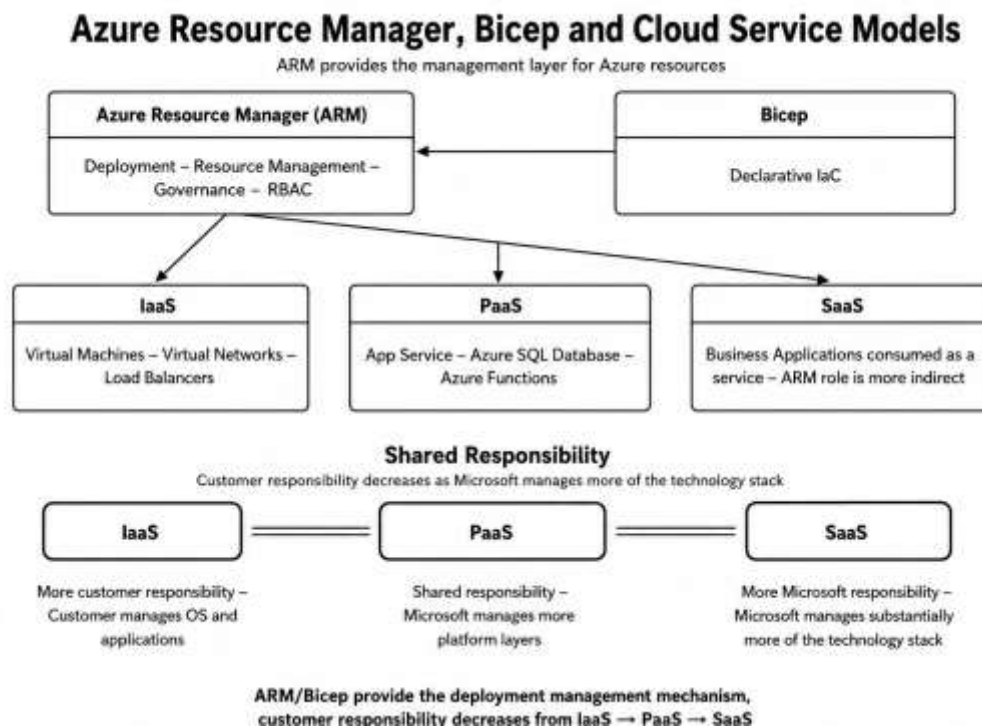


Figure 1. Azure Resource Manager, Bicep and Cloud Service Models

### 3. Deployment Types

Deployment choice should reflect organisational context rather than technological preference. Sector, operating countries, regulation, data sensitivity, budget, skills, growth and recovery objectives determine suitability. NIST defines four cloud deployment models: public, private,

community and hybrid. This report focuses on the public, private and hybrid models required by the brief (Mell and Grance, 2011).

Public cloud suits organisations requiring rapid scaling without major capital expenditure. In Azure, ARM supplies the management layer and Bicep declaratively defines proposed compute, storage, networking and PaaS resources (Microsoft, 2026a; Microsoft, 2026b). Risks include regional availability, overspending, data-egress charges, misconfiguration and lock-in. Budgets, policy controls and approved regions should be set before deployment.

Private cloud may be justified for banks or healthcare organisations where data sovereignty, latency or legacy integration outweigh higher hardware, staffing and recovery costs. It is not inherently more secure, and ARM and Bicep cannot manage every private-cloud platform. Azure Stack Hub provides an ARM-template-compatible option (Microsoft, 2026f).

Hybrid cloud can separate workloads by risk. A manufacturer may retain latency-sensitive factory systems on-premises while using Azure PaaS for analytics. Azure Arc can project supported non-Azure resources into ARM for selected governance and lifecycle capabilities (Microsoft, 2025). This improves consistency but adds identity, network, monitoring and recovery complexity.

No model is universally cheapest, safest or easiest. The organisation should document its rationale, apply least privilege and cost controls, and maintain an exit plan. Although Bicep is open source, it is Azure-specific (Microsoft, 2026b); migration may still require data transfer, IAM redesign, service refactoring and compliance revalidation.

*Table 1. Comparison of public, private and hybrid cloud deployment models*

| Factor                | Public cloud        | Private cloud              | Hybrid cloud          |
|-----------------------|---------------------|----------------------------|-----------------------|
| Scalability           | High                | More constrained           | Flexible              |
| Capital cost          | Lower               | Higher                     | Mixed                 |
| Management complexity | Lower               | Higher                     | Highest               |
| Data/control          | Less direct control | Greater control            | Workload dependent    |
| ARM/Bicep fit         | Strong              | Limited/platform dependent | Supported selectively |

## 4. Infrastructure Design

The proposed design combines IaaS and PaaS services within an Azure Virtual Network (VNet). For IaaS, two web server VMs behind an Azure Load Balancer provide redundancy if one becomes unavailable. Azure App Service provides a PaaS alternative, while Azure Storage supports files, blobs and backup data. Network Security Groups (NSGs) control permitted network traffic within the VNet, while Azure Monitor provides logs, metrics and alerts. Azure Monitor could also use machine-learning-based dynamic thresholds to identify unusual metric behaviour, providing AI-assisted anomaly detection without requiring fixed alert thresholds (Microsoft, 2026g). For regional resilience, UK South is primary, with UK West proposed for disaster recovery. However, adding redundancy, stronger security controls and multi-region recovery also increases cost and management complexity. Protection should therefore reflect workload importance, recovery requirements and budget. More expensive controls, such as



WAF, premium DDoS protection and full multi-region disaster recovery, can be included in the production design without necessarily being deployed in a small proof-of-concept. ARM provides the management and deployment layer, while Bicep defines the infrastructure as code. The PoC validated aspects of the design, with selected evidence in Appendix A and full evidence in the accompanying PoC document.

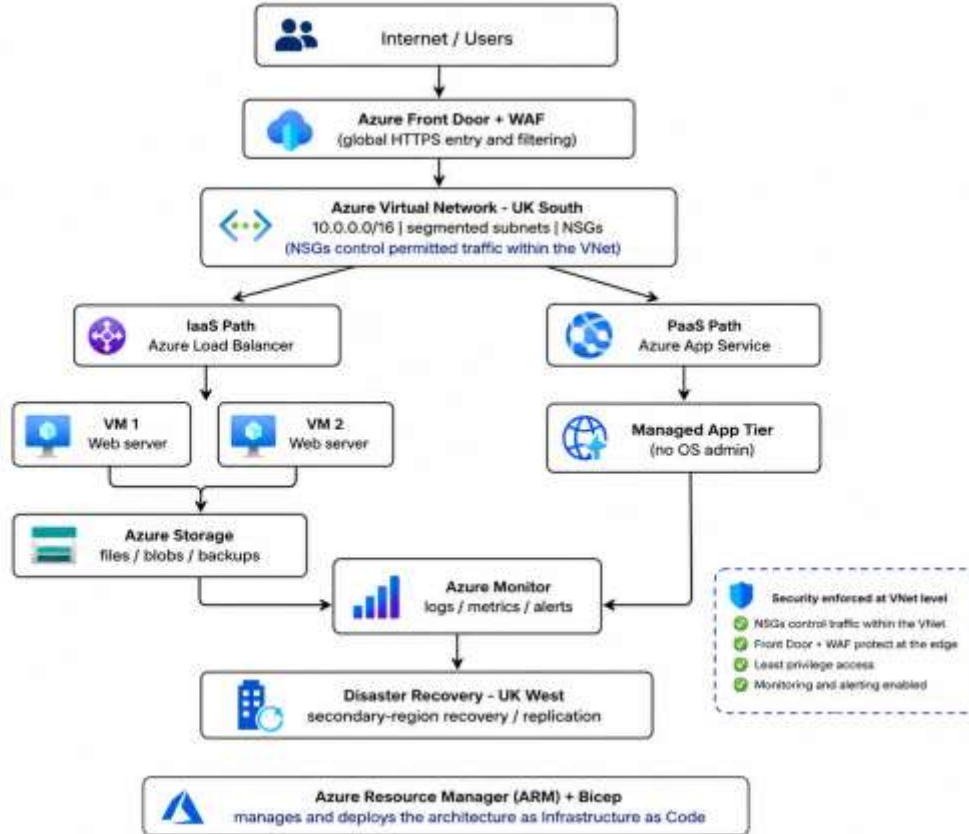


Figure 2. Proposed Azure infrastructure architecture using ARM and Bicep.

## 5. Real-World Example

A real-world ARM example comes from Microsoft's own IT organisation. Microsoft's Digital Cloud Engineering Services team had been using JSON-based ARM templates, but these did not always meet the organisation's security requirements. Teams therefore spent significant engineering time manually fixing security issues after deployment (Microsoft, 2024). This

approach aligns with the NIST Cybersecurity Framework's emphasis on integrating cybersecurity risk management into organisational processes (NIST, no date).

To address this, the team moved from JSON to Bicep and introduced role-based access control for template storage, alongside a CI/CD pipeline using Azure DevOps. This integrated security requirements into deployment rather than relying on post-deployment remediation (Microsoft, 2024).

The results were significant. Microsoft reported a 50 percent reduction in the engineering time required to develop templates, while the time previously spent fixing failed security controls was eliminated. More than 1,100 users also gained easier access to approved templates (Microsoft, 2024). Although Microsoft does not quantify the financial saving, reduced engineering and remediation time indicates potential operational cost savings, while implementing Bicep and CI/CD also requires skills, tooling and transition effort. These figures demonstrate improvements in both security and operational efficiency.

## **6. Conclusion**

Overall, ARM and Bicep provide a governed and repeatable approach to Azure deployment across different service and deployment models. Combined with appropriate resilience, monitoring and security controls, they can improve operational efficiency while supporting organisational requirements.

## 7. References

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## 8. Appendices

### 8.1 Appendix A – Bicep Proof-of-Concept Implementation Evidence

Selected evidence from the proof-of-concept is provided below. Full implementation evidence is supplied in the accompanying Azure Bicep PoC Implementation Evidence document.

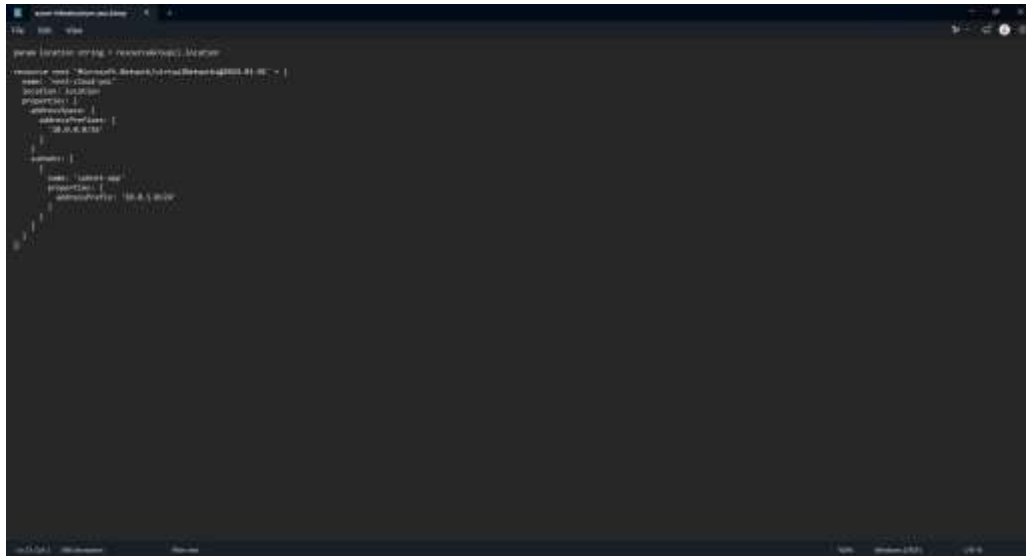
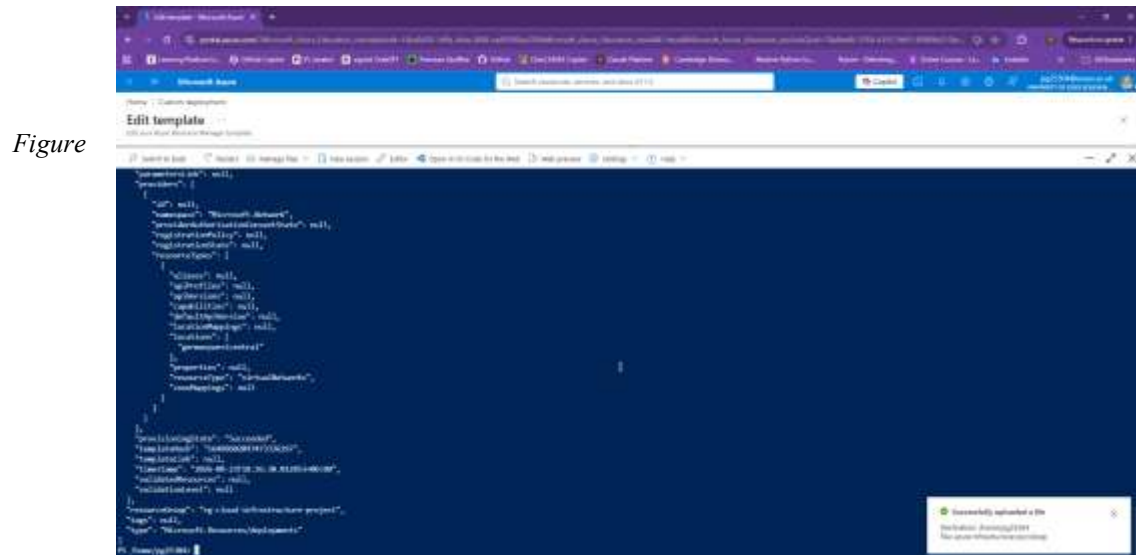


Figure A1. Bicep source defining the Azure Virtual Network and subnet used in the proof-of-concept.

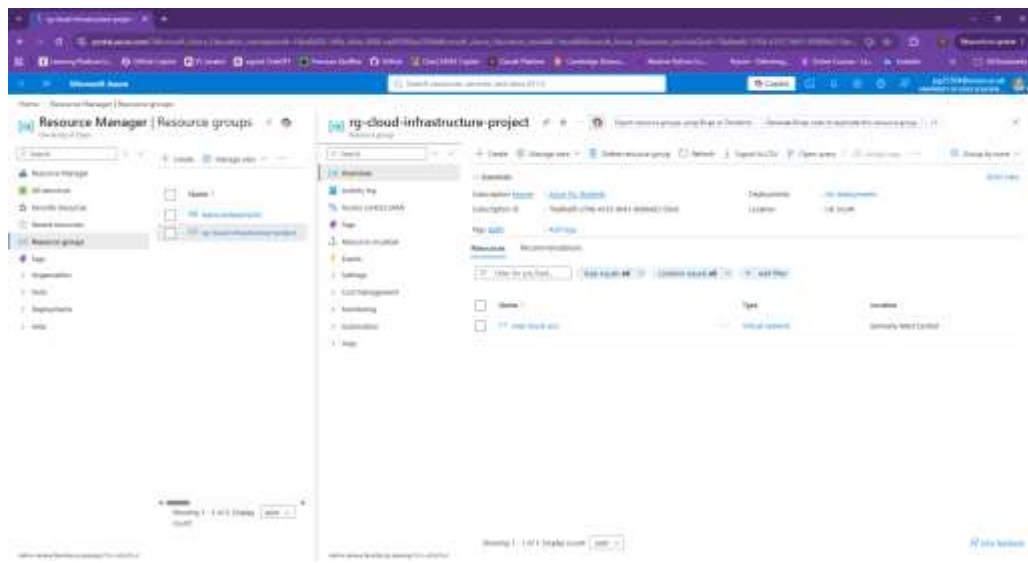


Figure

A2.

Successful Bicep deployment in Azure Cloud Shell, confirming a Succeeded provisioning state.

Figure  
Azure



A3.

Resource Manager showing the vnet-cloud-poc virtual network successfully provisioned in the project resource group.